



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2021-2022

TECHNOLOGY STUDIES (N550)

TIME: 1 HOUR 15 MINUTES

Component 1

Candidates answer on the Question Paper.
Additional materials: Standard drawing equipment

READ THESE INSTRUCTIONS FIRST

- Write your index number in the space provided above and on the A3 printed question sheet.
- Write in dark blue or black ink. Coloured and HB pencils may be used for diagrams/sketches and shading.
- Answer **ALL** questions.
- Check whether this assessment booklet consists of:
 - Questions **1 - 7** printed on **7** pages from page **2** to page **8**;
 - Question **8** printed on an **A3** sheet paper as an insert.
- Do not use correction fluid, highlighters or glue.
- All answers must be written in the spaces provided.
- The total marks for each question is given at the beginning of each question.
- The number of marks for each item is shown in brackets [].
- Question 8:**
 - All answers are to be drawn accurately using drawing instruments.
 - All construction lines **MUST** be shown.
 - All dimensions are in millimetres.
- The total number of marks for this paper is **50**.
- At the end of the examination, fold the A3 sheet and insert it in the assessment booklet.**

For Examiners' use										
Question No.	1	2	3	4	5	6	7	8	Total	Signature
Examiner										
Team Leader										
Quality Controller										
CE/ACE										

SECTION A**Question 1 (5 marks)**

Circle the correct answer. Each item carries one mark.

(i) Which eco-material is used to make baskets, mats and hats?

A Pandamus leaves

B Bamboo fibres

C Coconut coirs

D Palm leaves

(ii) The **tool** shown in Figure 1 is a wire

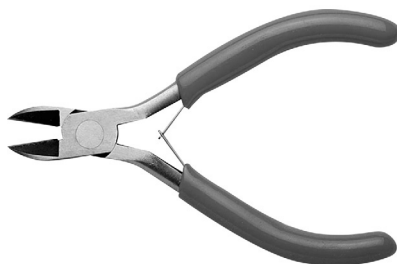


Figure 1

A stripper

B cutter

C heater

D bender

(iii) A pulley is used to transmit motion.

A linear

B oscillating

C reciprocating

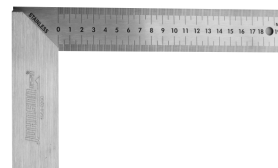
D rotary

(iv) Which **one** of the following is a holding tool?

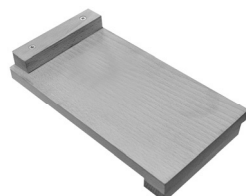
A



B



C



D



(v) In pneumatics, mechanical motion is achieved by using pressurised

.....

A air

B oil

C petrol

D water

Question 2 (5 marks)

Put a tick (✓) in the appropriate column to indicate whether the statements in the table below are True or False. The first one has been done for you.

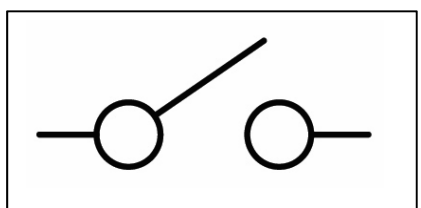
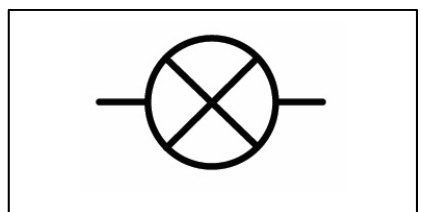
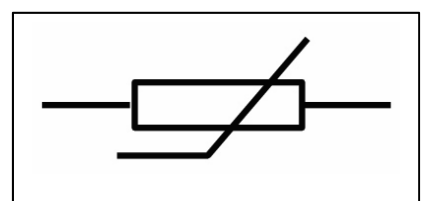
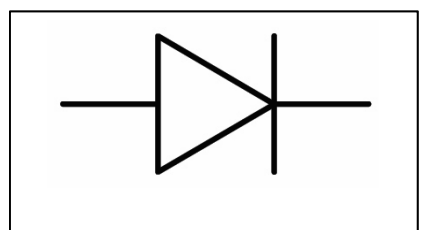
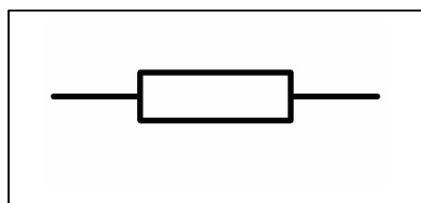
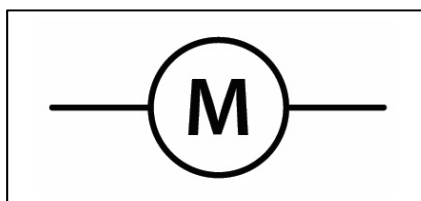
STATEMENT		TRUE	FALSE
Example: <i>Thermosetting plastics cannot be reshaped.</i>		✓	
a	A T-square is used to draw perpendicular lines.		
b	Tin is a malleable metal.		
c	Timber is classified as processed wood.		
d	Electrical insulators allow the flow of current.		
e	Eco-materials decay naturally.		

E
TL
QC
CE

E
TL
QC
CE

Question 3 (5 marks)

Match the symbols shown in Column A to their corresponding names in Column B.
An example is given.

COLUMN A**COLUMN B**

- Light bulb
- Thermistor
- Switch
- Electric motors
- Variable resistor
- Resistor
- Diode

Question 4 (5 marks)

Fill in the blanks with the appropriate word from the list given below. An example is given.

glass	Cedar	motion	alloy	specifications	design	turbines
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Example: Specifications are points that define the key features of the design solution.

- a) Red is used to make musical instruments.
- b) Green involves making products that cause no harm to the environment.
- c) Wind convert movement energy into electrical energy.
- d) Acrylic is often used as a replacement for as it is a transparent and tough material.
- e) Bevel Gear transmits at right angle.

Question 5 (5 marks)

- (a) **Figure 2** shows the line diagram of a lever used to lift a load. Label parts **A** and **B**.

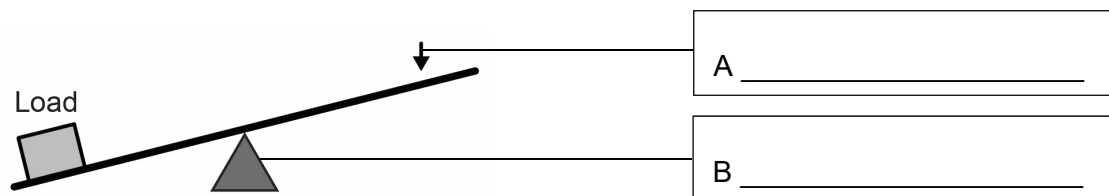
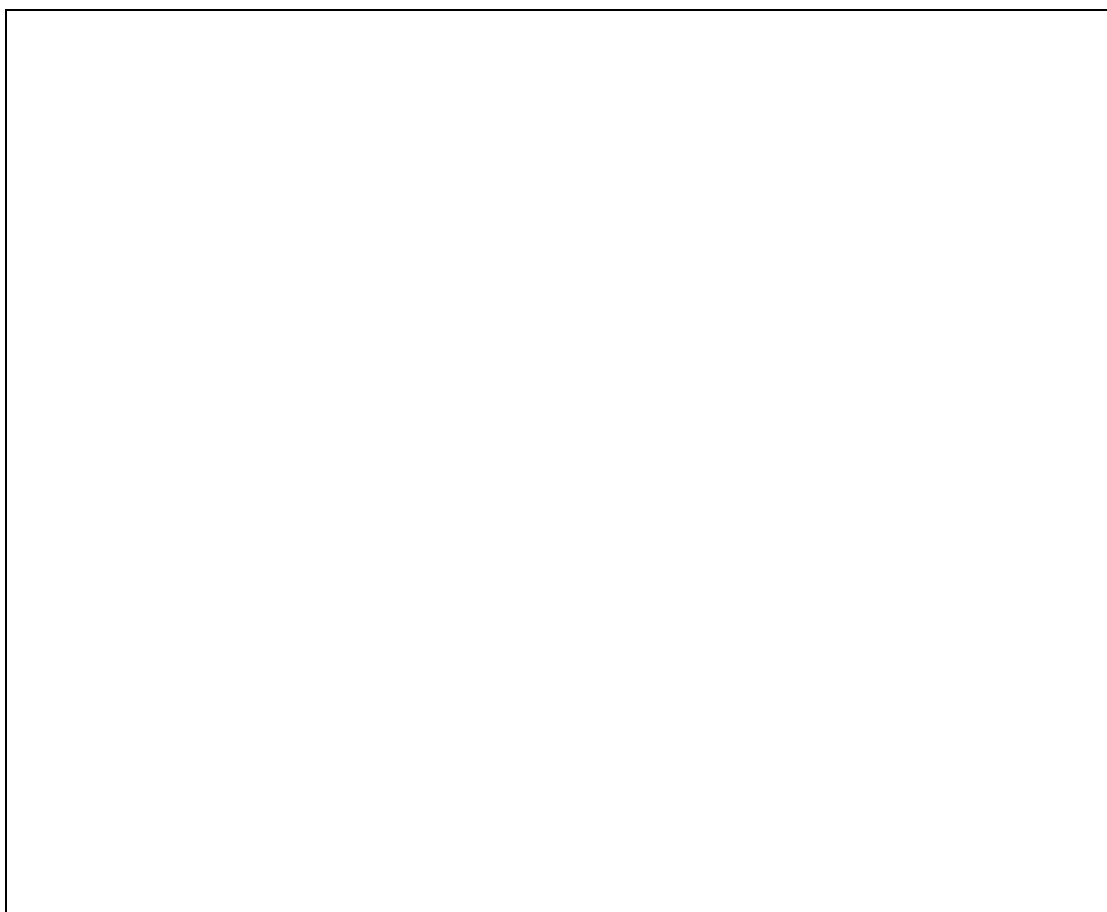


Figure 2: Lever symbol

[2]

- (b) Sketch a **wooden mallet** in the space provided below.



[3]

SECTION B**Question 6 (4 marks)**

Figure 3 shows an isometric view of a wooden shelf. Apply wooden surface texture rendering techniques using coloured pencils to enhance its appearance.

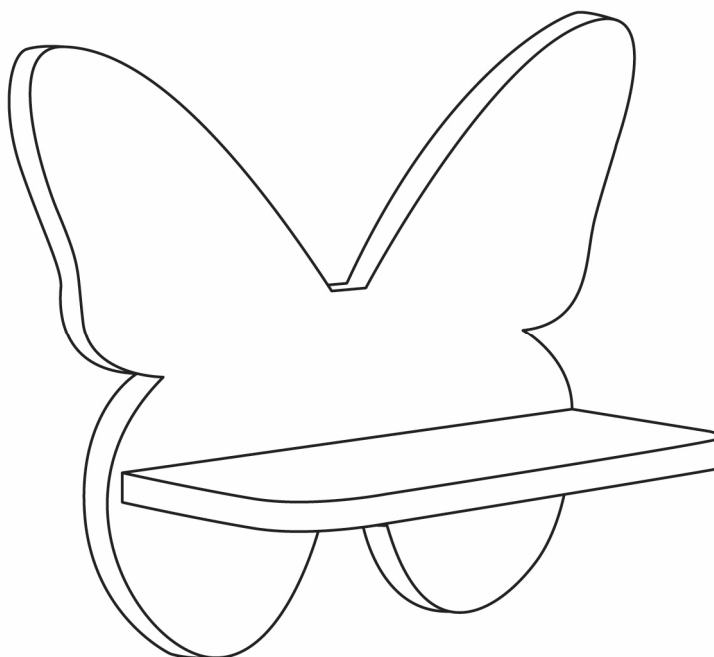


Figure 3: Isometric view of a wooden shelf

E
TL
QC
CE

Question 7 (6 marks)

Figure 4 shows three flower pots of different sizes. A display stand is required to hold the three flower pots in a living room.

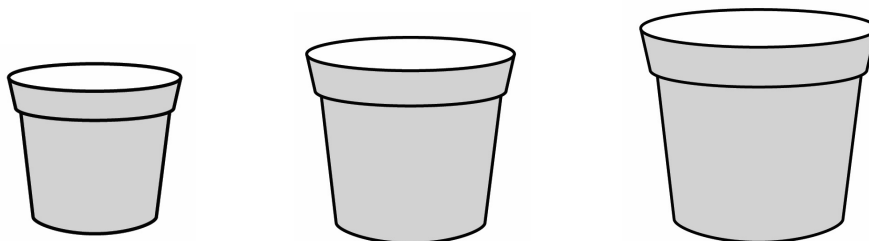


Figure 4: Set of three flower pots

In the space provided, sketch **one idea** for the display stand with an analysis of the function, material(s) used and safety.

Function:

Material(s) used:

Safety considerations:

E
TL
QC
CE